

1

Supplemental Materials: Simulation results

Supplemental Materials: Simulation results

To evaluate the finite sample properties of the f_{MACS} estimator, we perform two small scale simulations, one for continuous data and the other for categorical data.

Simulation 1: Continuous Indicators

In simulation 1, we simulate data from a unidimensional factor model with $p = 6$ items across conditions of (a) sample size ($n = 100, 250, 1000$ per group) and (b) number of groups ($G = 2$ or 4). For conditions with $G = 4$, we set α (latent mean) as $(0, -0.25, 0.25, 0.5)$ and ψ (latent variance) to $(1, 0.85, 1.15, 0.7)$. The first item was simulated to have noninvariant loading: for Group 1, $\lambda = (0.6, 1.0, 0.7, 0.8, 0.9, 0.7)$; for Group 2, the first loading is 0.8; for Group 3, the first loading is 1.0; and for Group 4, the first loading is 1.2. The second item was simulated to have noninvariant intercept: for Group 2 and Group 4, the intercepts are $\nu = (2.0, 3.0, 1.5, 3.5, 2.0, 3.0)$; for Group 1, the second intercept is 2.9, while for Group 3, the second intercept is 3.1. The uniqueness θ are drawn from uniform distributions with range $(1.0, 1.2, 0.8, 0.9, 1.0, 1.0) \pm 0.2$, resulting in composite reliability of ω between .78 and .82 for the groups. The population f_{MACS} values are 0.170 for Item 1 and 0.048 for Item 2.

For conditions with $G = 2$, the parameters for Group 1 are the same as in conditions with $G = 4$, while Group 2 has $\alpha = -0.25$, $\psi = 1.15$, and $\lambda_1 = 1.0$. Therefore, only one item has noninvariant loading, with $d_{\text{MACS}} = 0.301$.

For all conditions, we used *lavaan* (version 0.6-19) to fit a correctly specified partial invariance model with invariant constraints on all parameters for the invariant items, and no equality constraints for the noninvariant items. The sample models are all identified by setting the latent mean and variance to 0 and 1 for Group 1. We then computed the sample f_{MACS} or d_{MACS} values using the maximum likelihood estimates of the model parameters. In addition, we also computed bootstrap bias-corrected estimates of f_{MACS} (or d_{MACS}) as

$$\hat{f}_{\text{MACS}}^{bc} = \max(2\hat{f}_{\text{MACS}} - \bar{\hat{f}}_{\text{MACS}}^B, 0),$$

where $\hat{f}_{\text{MACS}}$ is maximum likelihood estimate of f_{MACS} and $\bar{\hat{f}}_{\text{MACS}}^B$ is the mean of estimated f_{MACS}

across $B = 249$ nonparametric bootstrap samples (see Davison & Hinkley, 1997; Efron, 1987).

Note that we chose a smaller number of bootstrap samples given the computational demand of the bootstrap; in practice, researchers should a larger B (e.g., 1,999).

We used 1,000 replications for each condition.

Results. As shown in Table 1, the sample f_{MACS} estimates are close to unbiased with $n = 250$, with the bootstrap bias-corrected estimates showing less bias than the naive estimates. Similar results were found for sample d_{MACS} in Table 2.

Simulation 2: Ordinal Indicators

In simulation 2, we used the same design in Simulation 1 with four groups, but with the items dichotomized. We first simulated continuous latent response variates for the indicators, y^* , using the same λ , α , and ψ as in simulation 1, but with $\theta = 1.0$ to resemble the theta parameterization for the ordinal factor model [citation]. We then compute y_{jg} as $I(y_{jg}^* > \tau_{jg})$ with threshold parameters $\tau = (0.5, -0.5, 1.0, 0.0, -2.0, 0)$ for Group 1. For the other three groups the thresholds are the same as in Group 1, except that $\tau_2 = -0.6$ and 0.4 for Groups 2, and 4, resulting in population f_{MACS} values of 0.099 and 0.044. *lavaan* is again used to fit a correctly specified partial invariance model with weighted least squares estimation and the theta parameterization. From this model, we used the sample estimates of λ and τ to compute f_{MACS} , as well as the bootstrap bias-corrected estimates.

Results. As shown in Table 3, the sample f_{MACS} estimates showed more bias with categorical indicators when, and the bootstrap bias-corrected estimates did not completely remove the bias with $n = 100$. The bootstrap bias-corrected estimates showed negligible bias when $n = 250$, while the naive estimates needed $n = 1,000$ to show negligible bias.

References

- Davison, A. C., & Hinkley, D. V. (1997). *Bootstrap Methods and Their Application*. Cambridge University Press.
- Efron, B. (1987). Better Bootstrap Confidence Intervals. *Journal of the American Statistical Association*, 82(397), 171–185. <https://doi.org/10.1080/01621459.1987.10478410>

Table 1*Bias of the sample f_{MACS} estimator in Simulation 1.*

n	Item 1 ($f_{\text{MACS}} = 0.170$)		Item 2 ($f_{\text{MACS}} = 0.048$)	
	Naive	BC	Naive	BC
100	.040 (.036)	.001 (-.001)	.037 (.035)	.012 (.010)
250	.017 (.017)	.002 (.002)	.017 (.016)	.003 (.003)
1,000	.004 (.004)	.000 (-.000)	.004 (.004)	-.000 (.000)

Note. BC = Bootstrap bias-corrected estimates. Bias values are presented as the mean (20% trimmed mean) across 1,000 replications.

Table 2

Bias of the sample δ_{fmacs} estimator in Simulation 1.

n	Naive	BC
100	.035 (.031)	-.001 (-.002)
250	.012 (.011)	-.001 (-.001)
1,000	.002 (.002)	-.001 (-.001)

Note. BC = Bootstrap bias-corrected estimates. Population $\delta_{\text{fmacs}} = 0.301$. Bias values are presented as the mean (20% trimmed mean) across 1,000 replications.

Table 3*Bias of the sample f_{MACS} estimator in Simulation 2.*

n	Item 1 ($f_{\text{MACS}} = 0.099$)		Item 2 ($f_{\text{MACS}} = 0.044$)	
	Naive	BC	Naive	BC
100	.083 (.080)	.029 (.025)	.068 (.061)	.020 (.007)
250	.037 (.036)	.006 (.005)	.018 (.016)	-.009 (-.012)
1,000	.010 (.010)	.001 (.001)	-.014 (-.014)	-.025 (-.027)

Note. BC = Bootstrap bias-corrected estimates. Bias values are presented as the mean (20% trimmed mean) across 1,000 replications.